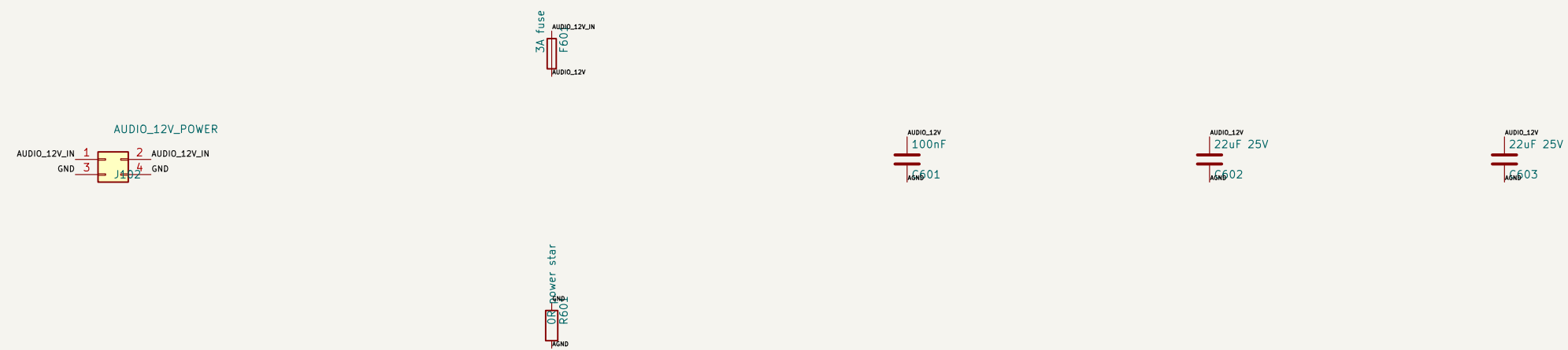


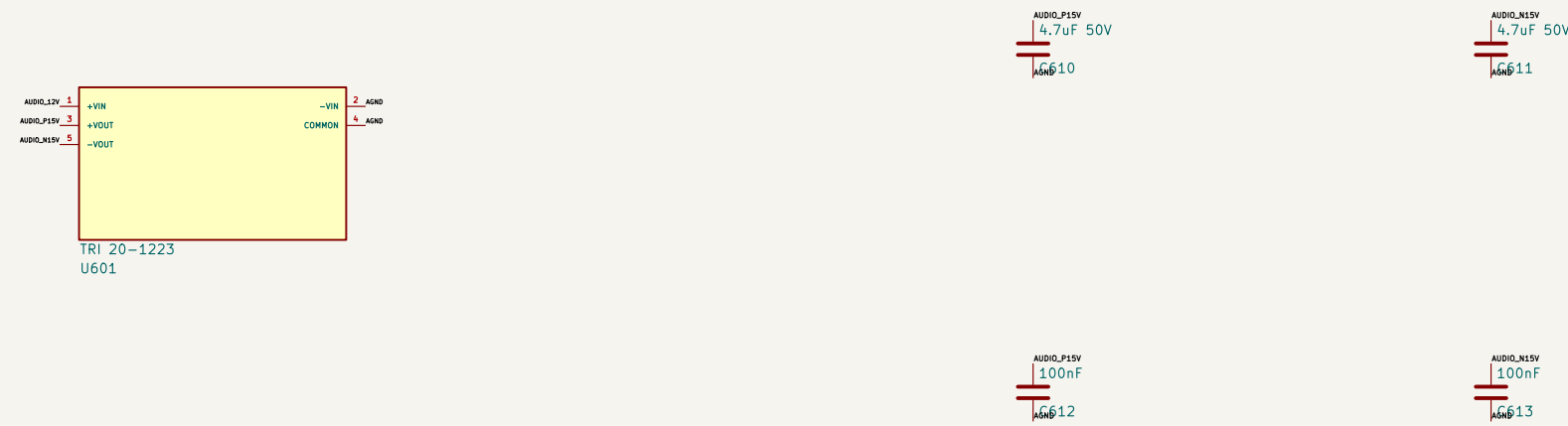
1. AUDIO_12V ENTRY AND SINGLE GROUND STAR



R606 is a wide 2512 zero-ohm start; no fan, RF, Ethernet, or chassis current may share the AGND slide

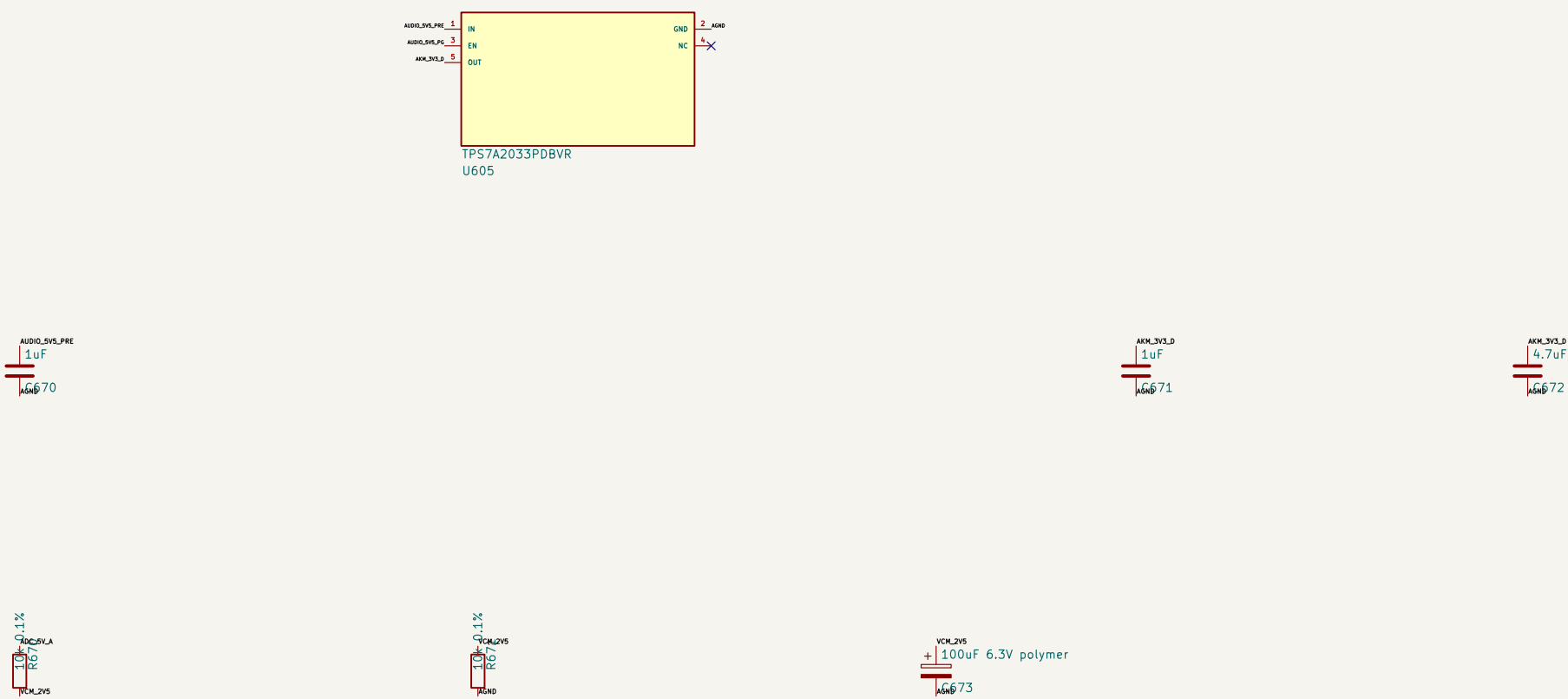
AUDIO_12V source is separately fused on CRS-CARRIER; F606 protects this removable board and harness locally.

2. ISOLATED BIPOLAR LINE-STAGE RAILS



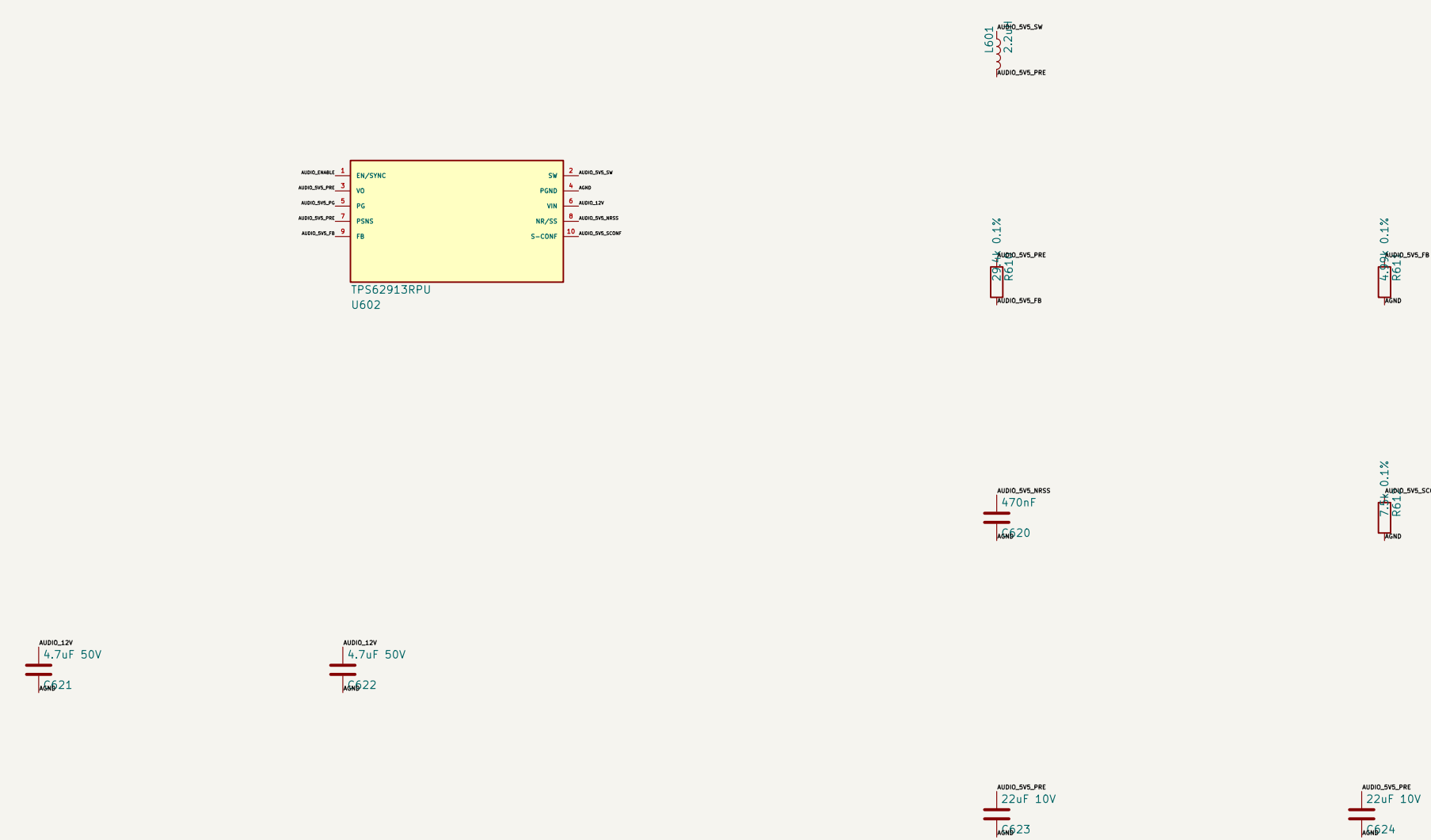
TRQ2: 9–18 V Input, ± 15 V at 670 mA per rail, 89% typical; place 4.7 μ F directly at each output pin.

5. AKM DIGITAL RAIL AND 2.5 V INPUT COMMON MODE



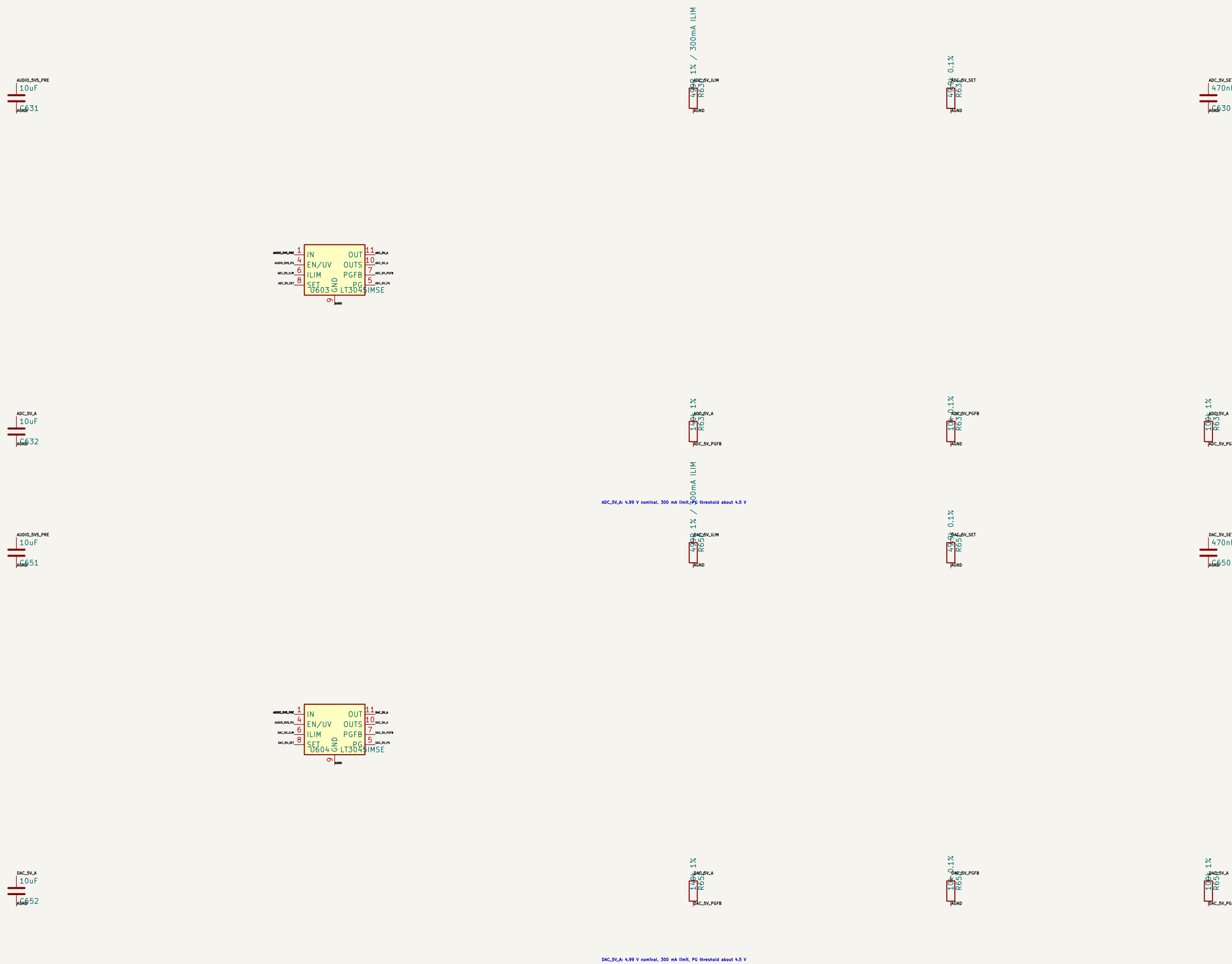
VCN2V5 is a quiet divider reference for the eight ADC input op-amps only; no external load is permitted.

3. QUIET 5.5 V PRE-RAIL



TPS62013: 2.2 MHz low-noise mode; 29.4k/4.99k feedback; 470 nF NP/SS; verify ripple below the AECM rail budget

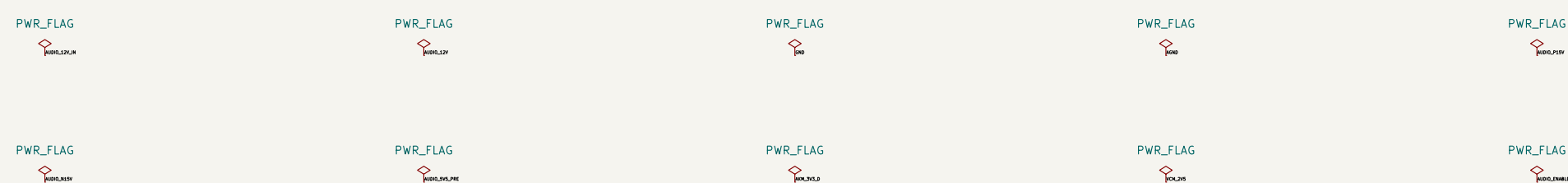
4. SEPARATE ADC AND DAC 5 V LT3045 RAILS



6. ERC SOURCES AND SEQUENCING CONTRACT

Release under AUDIO_ENABLE == YES/PC == (LTS445 PGs + AOM YES valid == clocks == resofs == SAC unmode == output relays

Any PG loss immediately de-energizes all output relays and reasserts OAC mode before software intervenes.



Power-ground connections: thermal rise, ripple, and conducted-emissions tests gate release
Separate LT3045 5 V/300 mA rail feed ADC and DAC; TPS7A2033 feeds AKM digital I/O
TRI 20-1223 creates +15 V; TPS62913 creates the quiet 5.5 V pre-rail
AUDIO12V is the only carrier power input; local 3 A fuse and one GND-to-AGND star
ProComm
Sheet: /
File: Audio-Power.kicad_sch

Title: ProComm AUDIO-8XB – Local Power, Sequencing, and Ground Star

Size: A1	Date: 2026-08-16	Rev: A1
KiCad E.D.A. 10.5.0-rc1		Ed: 1/1